

TEZPUR UNIVERSITY

End Term Examination, Autumn, 2023

BA301: Social Responsibility & Professional Ethics in Engineering

Full Marks: 60

Time: 3 hours

The figures in the right-hand margin indicate marks for the individual question.

1. Fill in the blanks

1×10=10

- a) Shortage of skilled labour leads to _____.
- b) The market where entry of new firms is difficult falls under _____.
- c) _____ theory studied both male and female perspectives of moral development.
- d) The third stage of value-based ethics is _____.
- e) _____ theory sees good as defined not by the action, but by the consequence.
- f) Contacting the legal counsel is a part of _____ appeal option.
- g) Price ceiling and _____ are some factors to consider while setting price.
- h) Doctors charging different fees to different patients for the same treatment is an example of _____ price discrimination.
- i) Patent is _____ in nature.
- j) If a product is found to be defective, then that product comes under product _____.

2. Write short answers:

2×7=14

- a) State any two characteristics of the market penetration price strategy.
- b) Why software piracy is ethically acceptable? (Mention any two reasons)
- c) What are the features of perfect competition? Mention any two.

d) What are the key challenges do you think startups face? (Any two).

e) Write any two software copyright protection techniques.

f) What are the probable sources (any two) of funds available for an entrepreneur?

g) What are the key areas (any two) to be assessed during the assessment stage of value based ethics?

1. Imagine you are an engineer working for a tech company that manufactures and sells a popular software product. The company is considering implementing a price discrimination strategy, where different prices will be charged based on factors such as user location, usage patterns, and the perceived value of the software to different customer segments. From an engineering ethics perspective, what considerations should the engineering team take into account when developing and implementing software that facilitates price discrimination? 6

4. What are the different types of risks an engineer face? Explain with appropriate examples. 10

5. Critically explain any one theories of moral development in ethics. 10

6. Please read the caselet carefully and answer the following questions. 10

In a quiet town, Emily and Olivia's lives were a tangled web of mistaken identities. Born on the same day, they grew up in different worlds as they were swapped at birth unknowingly. A routine DNA test revealed the switch, unraveling their destinies. Emily's mother, Mrs. Turner, faced a heart-wrenching decision. Struggling with the truth, she approached Olivia's parents, her voice quivering. "I can't bear to lose the daughter I've raised, but I can't ignore Olivia's right to know her true roots. Can we find a way for both girls to share their lives? Can I get my biological daughter back?" A tearful situation was reached, and the families embarked on an unconventional journey, challenging the conventional boundaries of motherhood.

a) How would you solve this conflicting situation (using the theories of ethics)? 5

b) Prepare a character table. 5